

Reviewer Pack — Minimal Hodge Decomposition Rank and Circuit Monotone Width ...

Entry 647c37275cee · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Hodge Decomposition Rank and Circuit Monotone Width Correlation via Tropicalization

Entry ID: 647c37275cee **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-02 06:04:09 UTC

1. Conjecture

Field A (mathematical branch): Tropical Geometry **Field B** (complexity object): Boolean Circuits

Statement:

For every d -regular Boolean circuit C , the minimal rank of its associated tropical Hodge decomposition ($\text{thd}(C)$) is linearly correlated with its monotone width $w_m(C)$, such that $\text{thd}(C) = \Theta(w_m(C))$.

Rationale (proposer's reasoning):

Tropical geometry provides a bridge between algebraic geometry and combinatorics. By tropicalizing Boolean circuits, we can potentially extract geometric information that correlates with circuit complexity measures like monotone width.

Taxonomy category: TROPICAL_GEOMETRY (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 729d4f63beb76320

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For every d -regular Boolean circuit, if the Pearson correlation coefficient between its minimal tropical Hodge decomposition rank and monotone width is greater than or equal to 0.7 for all seeds, then the conjecture is supported; otherwise, it is falsified.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - tropical geometry AND Boolean circuits AND Hodge decomposition - Boolean circuit AND monotone width AND tropicalization AND rank - minrank AND tropical Hodge decomposition AND correlation WITH monotone width

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def gaussian_elimination(A):
    n = len(A)
    for i in range(n):
        # Find pivot row
        max_row = i
        for j in range(i+1, n):
            if abs(A[j][i]) > abs(A[max_row][i]):
                max_row = j
        A[i], A[max_row] = A[max_row], A[i]

        # Eliminate below the pivot
        factor = Fraction(1, A[i][i])
        for j in range(i+1, n):
            factor_j = A[j][i] * factor
            for k in range(n):
                A[j][k] -= factor_j * A[i][k]

    # Back substitution to get RREF
    for i in range(n-1, -1, -1):
        for j in range(i+1, n):
            factor = A[j][i]
            for k in range(n):
                A[j][k] -= factor * A[i][k]
        A[i][i] = Fraction(1, A[i][i])

    return A

def rank(A):
    rref = gaussian_elimination(A)
    rank = 0
    for row in rref:
```

```

        if any(row):
            rank += 1
    return rank

def tropicalize(circuit):
    # Placeholder function to simulate tropicalization
    # Replace with actual tropicalization logic
    return circuit

def run_trial(seed: int) -> dict:
    random.seed(seed)

    n = random.randint(5, 40)
    d = random.randint(2, min(n-1, 3))

    # Generate a random d-regular Boolean circuit
    circuit = [[random.choice([0, 1]) for _ in range(d)] for _ in range(n)]

    thd_value = rank(tropicalize(circuit))
    wm_value = sum(1 for row in circuit if any(row)) # Placeholder for monotone width

    return {
        "metric_name": "correlation",
        "metric_value": thd_value * wm_value, # Simulated correlation
        "instances_tested": 1,
        "n_max": n,
        "conjecture_holds": False,
        "counterexample": "mapping_undefined"
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47]

    results = []
    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    mean_value = sum(result["metric_value"] for result in results) / len(results)
    std_value = math.sqrt(sum((result["metric_value"] - mean_value) ** 2 for result in results) / len(results))
    support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

    if all(result["conjecture_holds"] for result in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    else:
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample='mapping_undefined' first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4ed18800.py", line 85, in <module>
    trial_result = run_trial(seed)
                   ~~~~~^~~~~~

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4ed18800.py", line 67, in run_trial
    thd_value = rank(tropicalize(circuit))
                ~~~~~^~~~~~

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4ed18800.py", line 46, in rank
    rref = gaussian_elimination(A)
          ~~~~~^~~~~~

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4ed18800.py", line 33, in gaussian_elim
    A[j][k] -= factor_j * A[i][k]
    ~~~~^~~~
    ~~~~^^^
IndexError: list index out of range
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed during execution, which prevents us from verifying the conjecture's support conditions. | next: Investigate and fix the crash in the test code to allow for a proper verification of the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14031
2	preregistration	ollama_remote	glm4:latest	0	0	11125
3	novelty	ollama_remote	glm4:latest	0	0	8246
4	novelty	ollama_remote	glm4:latest	0	0	18268
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	28606
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	25540
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	20404
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	24221
9	judge	ollama_remote	glm4:latest	0	0	12735

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 163176 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in [research/audit/647c37275cee.jsonl](#))

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/647c37275cee.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/647c37275cee.tar.gz` (if generated)